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Propafenone versus Procainamide for Conversion of Atrial Fibrillation to Sinus Rhythm

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Summary

Background and hypothesis: Antiarrhythmic drugs are widely used for treatment of atrial fibrillation (AF) and restoration of sinus rhythm. This prospective, randomized, and controlled study compared the efficiency and safety of propafenone versus procainamide for the treatment of acute AF.

Methods: In all, 117 patients (55 women, 62 men, mean age 64.2 ± 13 years, median 63 years) who presented with AF were included in the study. Exclusion criteria were signs or symptoms of heart failure on physical examination, recent myocardial infarction or cardiac surgery, cardiogenic shock, or hypotension. Forty-one patients spontaneously recovered sinus rhythm; the remaining 76 patients were randomized to receive propafenone or procainamide. Propafenone was given at a dose of 2 mg/kg body weight intravenously (IV) over 30 min. Patients randomized to receive procainamide received a bolus of 100 mg IV administered every 5 min up to a maximum dose of 1 g. The clinical characteristics of the two groups were comparable.

Results: The number of patients who recovered sinus rhythm after the treatment was larger in the procainamide-treated group (Group 1) (69.5%) than in the propafenone-treated group (Group 2) (48.7%); $p < 0.05$. The time required for cardioversion was significantly lower in Group 1 (mean 4.1 ± 1.3 h), than in Group 2 patients (mean 7.3 ± 2.6 h) ($p < 0.01$).

Conclusion: In the present study, procainamide was more effective than propafenone for the treatment of AF of short duration.

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Introduction

Antiarrhythmic drugs remain the treatment of choice of atrial fibrillation (AF) for many physicians and patients. There is still a need, however, for clinical trials that evaluate the efficacy and the safety of the different drugs. Propafenone is a class 1C drug and was recently tested in a randomized, placebo-controlled study.[1]

Procainamide is the only intravenously (IV) administered class I antiarrhythmic agent available for clinical use in the U.S.[2] Its utility in the conversion of AF has been supported more by observation than by formal randomized comparison.[3,4] Intravenous procainamide is the treatment of choice for patients with the Wolff-Parkinson-White syndrome who have a preexcited ventricular response during AF if they are hemodynamically stable.[5]

The aim of this randomized study was to evaluate the efficacy and safety of propafenone compared with procainamide for the termination of acute AF.

Methods

Study Population

The study group included 117 consecutive patients (55 women and 62 men, mean age 64.2 ± 13 years, median 63 years) hospitalized with AF. Of these, 41 spontaneously recovered sinus rhythm. The remaining 76 patients were randomized to receive IV propafenone or procainamide.

Inclusion criteria for the study were AF, age >18 years and recent onset AF (lasting ≤ 2 weeks) and chronic AF (lasting > 2 weeks), either as a first or a recurrent episode. Recurrent AF was considered only if a prior episode was documented.

Patients were excluded from the study if they met one of the following criteria: signs or symptoms of heart failure on physical examination, recent myocardial infarction or cardiac surgery (< 3 months), cardiogenic shock or hypotension (systemic arterial pressure < 90 mmHg), electrocardiographic

(ECG) evidence of ventricular preexcitation, second- or third-degree atrioventricular block, previous diagnosis of sinus node disease, unstable hepatic or renal function, or evidence of digitalis intoxication and hypokalemia.

Patients were also excluded if they had been treated with amiodarone, if they were currently receiving treatment with antiarrhythmic drugs, digoxin, Ca antagonist, and beta blockers, or if they had a known allergy to one of the drugs.

Randomization

Patients who fulfilled the eligibility criteria were randomized 1:1 to procainamide (Group 1) or propafenone (Group 2) treatment. Patients enrollment was terminated when the difference between the two groups reached statistical significance. The study protocol was approved by the Ethical Committee of our university and informed consent was obtained from all participants.

Study Protocol

Before administration of the study drug, complete transthoracic and transesophageal echocardiograms were performed to evaluate the presence of cardiac disease and atrial thrombi.

The parameters measured were left ventricular diameters and kynesis, ejection fraction, and left atrial dimension. Complete medical history, physical examination, laboratory tests, coagulation parameters, 12-lead ECG, and chest roentgenogram were obtained at baseline.

Serial echocardiograms were performed after conversion to sinus rhythm.

Pharmacologic Conversion

At admission, patients were hospitalized in the ECG telemetry ward. Transesophageal echocardiography was performed in all patients with AF lasting > 48 h. Anticoagulation therapy to prevent thromboembolic events was administered according to standard care. Patients who had AF of unknown duration or for > 48 h received anticoagulants for approximately 3 weeks before and 4 weeks after cardioversion.[6] Patients who had negative transesophageal examination were treated with short-term anticoagulation (IV heparin for 48 h before cardioversion and then anticoagulants for 4 weeks).[7]

Patients in Group 1 received a bolus of procainamide of 100 mg IV administered every 5 min up to a maximum dose of 1 g.[3] Patients in Group 2 received 2 mg/kg body weight of propafenone dissolved in 100 ml of 5% glucose over 30 min.[1]

Heart rhythm was continuously monitored during and after drug administration. Conversion to sinus rhythm was confirmed by a 12-lead ECG recorded within 24 to 48 h after IV drug administration. Patients who did not convert to sinus rhythm within 48 h after pharmacologic therapy underwent DC shock.

Patients were treated with 250 mg of hydroquinidine retard 2 times/day after the restoration to sinus rhythm.

Statistical Analysis

The primary objective was to evaluate the efficacy of IV propafenone compared with IV procainamide for conversion to sinus rhythm. The efficacy outcome was tested by means of a chi-square test.

Data are expressed as mean value ± 1 standard deviation. A discriminant analysis was performed. A p value < 0.05 was considered significant.

Results

In all, 117 patients were enrolled in the study. Underlying cardiac disease was examined. In 70 patients (60%), no underlying cardiac disease was present. The duration of AF in the total population was < 2 weeks in 60 patients (51.3%) and > 2 weeks in 57 patients (48.7%). A prior episode of AF had been documented in 47 patients (40%).

The clinical characteristics of the two groups were comparable (Table I).

Pharmacologic Conversion

Before IV therapy, 41 patients spontaneously converted to sinus rhythm. Of the remaining 76 patients, 38 patients were randomized to receive procainamide (Group 1) and 38 were randomized to receive propafenone (Group 2).

The number of patients who recovered sinus rhythm after IV treatment was larger in Group 1 (69.5%) than in Group 2 (48.7%); $p < 0.05$.

TABLE I Demographics and clinical characteristics of patients at time of enrollment

	Propafenone (38 patients)	Procainamide (38 patients)	p Value
Demographics			
Age (years) (mean ± SD)	64.3 ± 12	63 ± 13	NS
Male	26	29	NS
Body mass index	25.6	26.5	NS
AF pattern			
Recent onset (< 2 weeks) (%)	30 (79)	32 (84)	NS
Chronic (> 2 weeks) (%)	8 (21)	6 (16)	NS
Prior episodes of AF	19	15	NS
Clinical characteristics			
Coronary artery disease (%)	4 (10.5)	5 (13)	NS
Hypertension (%)	6 (16)	5 (13)	NS
No structural heart disease (%)	23 (60.5)	22 (58)	NS
Others (%)	5 (13)	6 (16)	NS

Abbreviations: AF = atrial fibrillation, SD = standard deviation.

The conversion rate based on left atrial size and duration of AF is shown in Table II. No significant difference was observed between the two groups. The duration of AF was < 2 weeks in 30 Group 2 patients (79%) and in 32 Group 1 patients (84%).

The time to conversion to sinus rhythm was also analyzed. Group 1 patients recovered sinus rhythm in a mean of 4.1 ± 1.3 h, while Group 2 patients recovered sinus rhythm in a mean of 7.3 ± 2.6 h ($p < 0.01$).

Direct current shock restored sinus rhythm in 78% of the patients who did not recover sinus rhythm after drug treatment, and was equally effective ($p > 0.5$) in Group 1 (76%) and Group 2 (78%).

Echocardiographic features were similar in the two groups at baseline. The left atrial dimension of patients in both groups appeared larger in those who did not recover sinus rhythm.

Mean baseline heart rate and blood pressure were similar in the two groups and did not influence the rate of conversion.

TABLE II Conversion rates after drug therapy

	Propafenone		Procainamide	
	AF	SR	AF	SR
Left atrium < 40 mm	12	11	5	16
Left atrium > 40 mm	8	7	7	10
Recent onset AF				
< 2 weeks	5	14	3	19
Chronic AF				
> 2weeks	15	4	9	7
All	20	18	12	26

Abbreviations: AF = atrial fibrillation, SR = sinus rhythm.

Tolerability

Throughout the study, drug-related side effects were reported in four Group 2 patients (10%) and in three Group 1 patients (7.8%; p = NS). One patient in Group 1 developed a sinus bradycardia, one patient had temporary hypotension, and one patient developed allergic dermatitis. One patient in Group 2 developed severe junctional bradycardia, but recovered completely and recovered sinus rhythm in a few minutes. Two patients had gastrointestinal upset and one patient developed hypotension a few minutes after the end of infusion. No deaths were observed.

Discussion

Atrial fibrillation is the most common arrhythmic disorder. It is often associated with symptoms and usually responds poorly to pharmacologic therapy.[8] Cardioversion of AF to sinus rhythm is associated with an improvement in the hemodynamic status and a decrease in the risk of thromboembolic events.[9, 10] In recent onset AF, spontaneous cardioversion to normal sinus rhythm frequently occurs within 12 h in patients receiving an atrioventricular nodal slowing agent.[2] However, for a more rapid cardioversion or for AF of longer duration, a class IA, class IC, or class III antiarrhythmic drug is required.[2] Procainamide is an IV class I antiarrhythmic drug the conversion effect of which is dependent on arrhythmia duration.[3] Propafenone hydrochloride is a class IC antiarrhythmic drug that can be used either IV or orally. For conversion of AF, IV administration can be followed by an oral dose for prophylaxis. The speed of infusion may influence the rate of cardioversion: slow infusion (> 30 min instead of 10 min) may be less effective because sufficiently high tissue levels are not obtained.[1]

In the present study, the efficacy of IV propafenone and IV procainamide was compared in the cardioversion of AF. Sinus rhythm was restored in 69.5% of Group 1 patients compared with 48.7% of Group 2 patients. A previous study using procainamide showed a conversion rate of 58%, and the conversion was dependent upon the duration of the arrhythmia.[3] Comparisons between procainamide and other antiarrhythmic drugs have been made previously,[4, 11] but very few studies compared procainamide and propafenone.[12] Boahene *et al.*[12] evaluated the effects of IV procainamide and propafenone in 55 patients with acute AF and Wolff-Parkinson-White syndrome. They suggest that IV propafenone is superior to procainamide in that it immediately slows the ventricular response of AF, although procainamide may be more effective in terminating the arrhythmia.[12]

One study compared seven different drugs, including procainamide and propafenone,[13] but because of the small sample size the results of individual agents at 3 h were overlapping.

A multicenter study comparing the conversion rate of propafenone, propafenone plus digoxin, and quinidine or placebo showed that the most successful regimen was the combination of propafenone and digoxin, with a conversion rate of 82% at 12 h.[14] The recently published Propafenone Atrial Fibrillation Trial, comparing IV propafenone with placebo, reported a conversion rate of 29% in the treatment group.[1]

We observed higher conversion rates with both drugs in the present study. Two factors may have influenced the results of the present study—the high number of patients with lone AF and the short duration of AF, lasting < 2 weeks in 70% of the patients.

The time to conversion was also shorter in patients treated with procainamide than in patients treated with propafenone.

Upon analysis of the echocardiograms obtained from patients, it was evident that patients who did not recover sinus rhythm after drug therapy had a more dilated left atrium than patients who responded to the therapy. The dimension of the left atrium was similar in the patients cardioverted using procainamide or propafenone. It is well known that left atrial dimension influences the recovery and the maintenance of sinus rhythm.[15, 16] Mean baseline heart rate and blood pressure were similar in the two groups of patients. There was no difference in these parameters between patients who recovered sinus rhythm after pharmacologic treatment with either drug. No relationship was derived between sympathetic and nonsympathetic AF and response to the different drugs.

Therapy with propafenone and procainamide was associated with a relatively low incidence of side effects. This is a positive attribute of these drugs compared with other drugs that have shown a higher incidence of adverse events.[17, 18]

It is well known that arrhythmic events can occur in patients with recent myocardial infarction or left ventricular dysfunction due to the proarrhythmic effect. For this reason, patients with previous myocardial infarction and reduced ejection fraction were excluded from the study. This can be considered to be a limitation of the present study.

Conclusion

Our results showed that procainamide is more effective than propafenone in the cardioversion of AF of short duration, lasting less than 2 weeks. Procainamide is the drug of choice in this group of patients, because it had a higher incidence of cardioversion to sinus rhythm and a lower incidence of adverse effects.

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